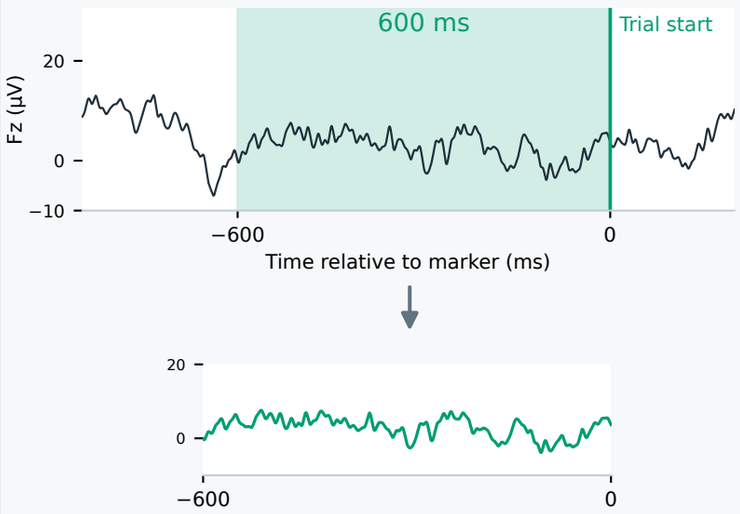
Fixation cross → Trial starts → encoding → Full sequence + choices → Response

Trial phases shown schematically; durations are not to scale.

Pre-trial rest EEG

Before the trial-start marker

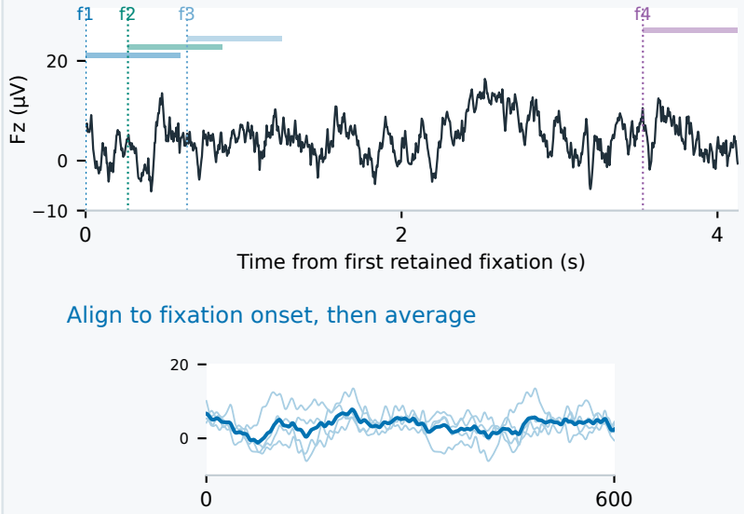


One pre-trial segment per trial

Fixation cross displayed before trial onset

Fixation-related EEG (FRP)

After each retained sequence fixation



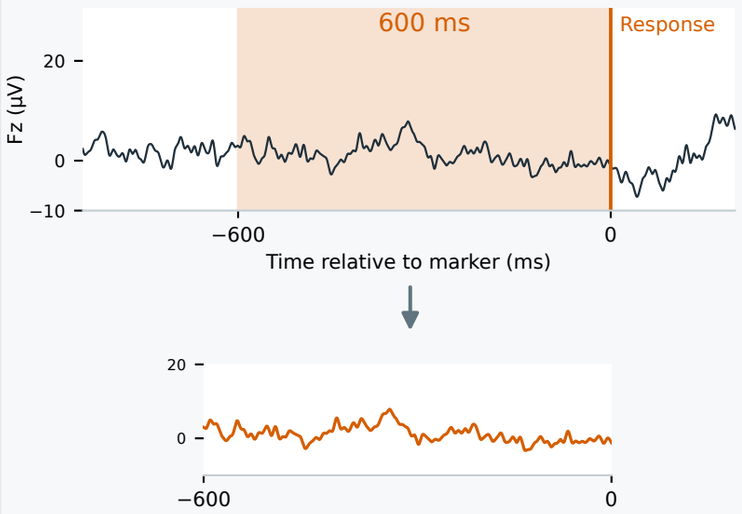
Align to fixation onset, then average

One trial FRP from multiple epochs

Sequence fixations ≥100 ms; windows may overlap

Response-locked EEG

Before the response marker



One pre-response segment per trial

Omissions: end of the response window

Shared construction of representational geometry

Regional time courses
Average electrodes at each time point

→

8 pattern vectors
Average trials within pattern + session

→

Session RDM
Pairwise correlation distance: 1 − r

→

Participant → group
Average RDMs within, then across people

Repeat separately for each EEG measure and each of five electrode groups: frontal, temporal, central, occipital and parietal.

Same 600-ms duration does not imply equivalent measurements.

FRP averages multiple post-fixation windows; response and rest use one window per trial. No fixation-specific baseline correction is applied to FRPs.

Measured EEG: participant 1, session 1, trial 1, electrode Fz; four retained sequence fixations. The trial-phase diagram above is schematic.